

THE ABERDARE TECHNICAL COLLEGE

ICT TECHNICIAN LEVEL 5

CONTINUOUS ASSESSMENT TEST (CAT) 2

TERM 3

UNIT: COMPUTER SOFTWARE
TIME: 1 HOUR 30 MINUTES
TOTAL MARKS: 50 MARKS

92%

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INSTRUCTIONS

- ✓ Answer ALL questions.
- ✓ Provide clear and technical explanations where required.
- ✓ Use relevant examples where necessary.
- ✓ This paper focuses on software testing and Postman tool usage.

Questions

1. Explain the importance of software testing in ensuring software functionality. (4 Marks)
2. State FOUR reasons why software testing is important in system development. (4 Marks)
3. Differentiate between functional testing and non-functional testing. (4 Marks)
4. State THREE software testing techniques used in verifying software functionality. (3 Marks)
5. Explain TWO advantages of using automated testing over manual testing. (3 Marks)
6. Describe the role of the following software testing tools:
 - a) Test Complete (2 Marks)
 - b) Selenium (2 Marks)
 - c) Postman (2 Marks)
7. Explain what Postman is and its use in software testing. (3 Marks)
8. Describe step-by-step how Postman is used to test an API. (5 Marks)
9. State FOUR HTTP methods used in Postman and briefly explain each. (4 Marks)
10. Explain TWO benefits of API testing using Postman. (3 Marks)
1. Describe the steps involved in testing a new software application before deployment. (4 Marks)
2. State THREE common errors detected during software functionality testing. (3 Marks)
3. Explain TWO challenges faced during software testing and how they can be solved. (3 Marks)
4. Describe how regression testing helps maintain software functionality after changes. (3 Marks)
5. Explain the role of test cases in software functionality testing. (3 Marks)

1. Importance of Software testing

- Help choose application with certainty ✓
- Help find bugs ✓
- Improves the software quality ✓
- Help prevent costly errors ✓

2. Detects bugs early ✓

- Ensures the software meets user requirements ✓
- Improve software quality and reliability ✓
- Reduces maintenance cost after development ✓

3. Functional testing → tests what the system does according to requirements ✓

Non functional testing test how system performs ✓

1. Black box testing ✓

White box testing ✓

Grey box testing ✓

5. Faster execution ✓

Since automated tests run quickly and repeatedly and consistently - some test are repeated and reduce human errors ✓

6. Define the tool ✓

Selenium → It is used for automating web browser testing ✓

Test complete → Test functional web and mobile applications ✓

Postman → Used for testing APIs ✓

7. Postman is an API testing tool ✓

Used to send HTTP request to APIs ✓

8. Install app and fire ~~request~~ ~~of~~ ~~Postman~~
Create a new request ~~of~~ ~~choice~~ ✓
Enter API URL and Send OS
Check response ✓
Save request in a collection for reuse.

9. GET → retrieve data from server.
POST → Send new data to server. ✓
PUT → Update existing data. ✓
DELETE → Remove existing data from server.

10. Easy and fast testing of backend logic without UI

11. Requirement analysis ✓
Test Planning and test case creation ✓
Execute test ✓
Fix bugs and ~~retest~~ regression testing ✓
Final validation and approval ✓

12. Functional bugs like

User interface issues ✓
Wrong input handling ✓

Tight deadlines ✓ Use of automation and prioritize critical tests
lack of requirements clarity ✓ Have earlier and explanatory communication with the user.

Regression testing reruns previous test cases after code changes to ensure that new updates have not broken existing functionality.

15. ~~Things~~

It help ensure consistency ✓

It help in software completion ✓

Help in repeatable testing ✓